Detecting Malignant Skin Lesions from Image and Clinical Data

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This solo project classified skin lesions as needing treatment or not, combining smartphone images with patient clinical metadata from PAD-UFES-20 [2]. Much clinical metadata was missing (patients skipped fields, or curators removed unverifiable entries), so handling missingness well was a core part of the task. The project had two parts: build a random forest (RF) model on engineered image features, then design a new model that outperforms it on held-out test data.

RF Model

I engineered three image features: mean color intensity and color variance (all channels/pixels, capturing pigmentation and color irregularity) and edge density (grayscale gradient-magnitude thresholding, capturing border complexity), merged with the tabular data by `img_id`. Numeric fields were median-imputed; binary fields were normalized to true/false/missing and one-hot encoded; other missing values were coded as an explicit "Missing" category rather than dropped, since absence itself was informative (unverified $\neq$ negative). The training set was heavily imbalanced (BCC alone was 57%), so I augmented (flip, rotate, mild brightness/contrast) all classes except BCC, scaling copies to each class's rarity (e.g., 6× for MEL/SEK, 0× for BCC) to boost minority representation without fully equalizing class sizes.

Cross-validation used `StratifiedGroupKFold` on patient ID to prevent leakage. I fixed per-class decision thresholds on a held-out split ($\tau = 0.1$ for MEL, $\tau = 0.25$ for NEV, since missing a malignant MEL is far costlier than over-flagging a benign NEV), then ran a 5-fold hyperparameter search under those fixed thresholds, optimizing *minimum age-group* macro AUROC so the model couldn't win by only fitting the dominant class or age group. The final model (`n_estimator`=600, `max_depth`=15, `max_features`=$\sqrt{\cdot}$, `min_samples_split`=12, `min_samples_leaf`=5) reached a test AUROC of 0.901, with high recall (0.996) but low specificity (0.304): it rarely misses a dangerous lesion but produces more false alarms, an acceptable trade-off for a treatment-triage tool.

Open-Ended Challenge

For Part 2, I moved to a fusion model: a frozen, pretrained EfficientNet-B0 (small parameter count, to limit overfitting on a small dataset) producing a 256-d image embedding, concatenated with an MLP embedding of the tabular features and fed to a classification head. I also re-engineered the image features using the dermatology ABCDE framework [1]: per-channel color entropy (replacing a single mean-color scalar) and diameter-based features (mean diameter, diameter >6mm). An ablation study showed diameter features paired with edge density gave the best minimum age-group AUROC, so the final feature set was diameter mean, diameter >6mm, and edge density.

I also improved sampling: instead of shuffled fine-label batches, I used a `WeightedRandomSampler` (with replacement) over coarse labels to oversample rare age×class combinations, additionally oversampling the negative senior group 1.5× after the model over-predicted negative there. This raised mean minimum age-group AUROC to 0.856 (±0.066) across folds. Early stopping (patience=3) prevented overfitting past peak validation. A configuration with slightly higher validation AUROC was rejected for being far less stable across folds; the final model (lr=0.001, weight decay=0.0001, dropout=0.3, batch size=16) reached a test AUROC of 0.863 and minimum age-group AUROC of 0.797.

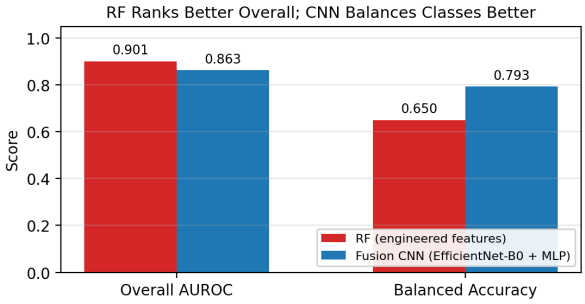


Figure 1: RF ranks lesions needing treatment better overall, but the CNN's balanced accuracy is substantially higher, thanks to weighted sampling of underrepresented subgroups. Both models struggled most with uniformly red, low-opacity lesions, suggesting neither feature set fully captures that cue.

References

[1] American Academy of Dermatology. What to look for: ABCDEs of melanoma. `https://www.aad.org/public/diseases/skin-cancer/find/at-risk/abcdes`, 2026. Accessed July 4, 2026.

[2] Andre G.C. Pacheco, Gustavo R. Lima, Amanda S. Salomão, Breno Krohling, Igor P. Biral, Gabriel G. de Angelo, et al. Pad-ufes-20: A skin lesion dataset composed of patient data and clinical images collected from smartphones. *Data in Brief*, 32:106221, 2020.